DreamGuard - Checkpoint 1

Andrew Scouten
yzb2@txstate.edu
Texas State University

Avery R Vanausdal
arv107@txstate.edu
Texas State University

Research Question

Does replacing the standard guardian grid with an aesthetically-integrated passthrough blend reduce cybersickness and presence disruption without degrading safety outcomes?

Current Progress

The DreamGuard prototype is running on a Meta Quest 3 in a Godot 4 VR dungeon environment. At this stage, two core components are functional: a basic dungeon scene and a basic windowed passthrough overlay. The dungeon scene provides a representative VR environment for testing safety transitions (Figure 2). The passthrough component uses spatial shader-based compositing to cut a rectangular window into the scene, revealing the physical environment through the headset's cameras. A menu system allows toggling between passthrough configurations during a session (Figure 1). All other planned features—multiple blending styles, automated trigger logic, study instrumentation—remain unimplemented and are the focus of the next checkpoints.

Key Challenges

Three technical obstacles were encountered while reaching the current prototype state: (1) GPU culling silently discarded the fullscreen passthrough quads; (2) the passthrough node unconditionally reset `transparent_bg` to `false` on deactivation, breaking the compositing pipeline; and (3) Godot 4 Forward Mobile's multiview `SCREEN_UV` maps to the combined stereo render target rather than each eye independently, causing the passthrough window to appear offset per eye and producing double vision. The third issue has not yet been fully resolved and is under active investigation.

Plan for Next Checkpoints

With a basic windowed passthrough and dungeon scene in place, the next checkpoint will focus on: (1) resolving the per-eye double-vision rendering issue and validating the fix on device; (2) implementing additional passthrough blending styles beyond the basic window (e.g., vignette, animated fade) so that conditions can be meaningfully compared; (3) adding trigger logic and session instrumentation (logging of trigger events, blend intensity over time, and active style condition); and (4) finalizing the study design..

Figure 1: In-headset menu for toggling between passthrough configurations.

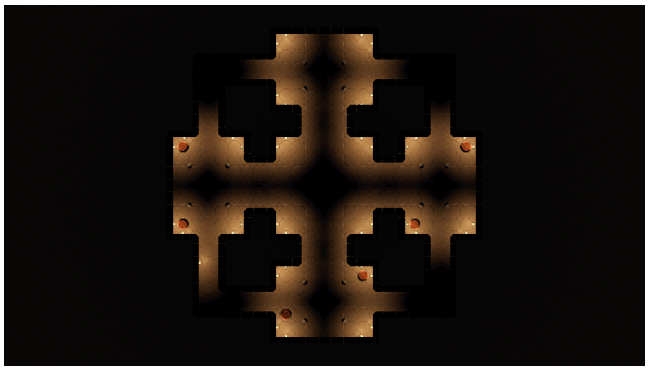

Figure 2: Top-down view of the dungeon scene used as the VR test environment.

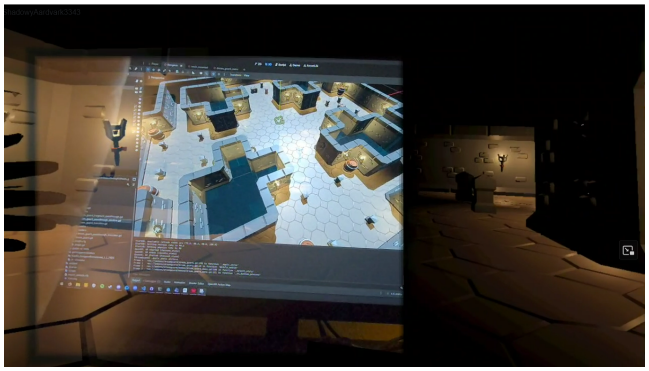

Figure 3: Windowed passthrough as seen in-headset.